

Heterogeneity in the US gender wage gap

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Abstract

As a measure of gender inequality, the gender wage gap has come to play an important role both in academic research and the public debate. In 2016, the majority of full-time employed women in the United States earned significantly less than comparable men. The extent to which women were affected by gender inequality in earnings, however, depended greatly on socio-economic characteristics, such as marital status or educational attainment. In this paper, we analyse data from the 2016 American Community Survey using a high-dimensional wage regression and applying double lasso to quantify heterogeneity in the gender wage gap. We find that the wage gap varied substantially across women and that the magnitude of the gap varied primarily by marital status, having children at home, race, occupation, industry, and educational attainment. These insights are helpful in designing policies that can reduce discrimination and unequal pay more effectively.

Keywords: discrimination, gender wage gap, heterogeneity, high-dimensional models, lasso

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1 Introduction

Empirical studies on the gender wage gap have come to play an important role in the public and academic debate. Most studies that have attempted to quantify gender inequality in earnings to date employ decomposition methods (Blinder, 1973; Oaxaca, 1973) and, as a key result, report an unadjusted and an adjusted estimate of the gender wage gap. Using recent data from the 2016 American Community Survey (ACS), the unadjusted gender wage gap¹ amounts to around 24% for full-time, year-round employed women with a high school degree or lower level of education. For employees with higher educational attainment, the unadjusted gap is approximately 33%. This unconditional measure of gender-based inequality in earnings is often used in political discussion but does not account for differences in observational variables between men and women. For example, men often choose industries and college majors that lead to higher wages on average. Hence, controlling for variables is key and adjusting for differences in observable characteristics changes the pay gap to approximately 17% (for women with high school degree or lower) and 14% (for women with at least a college degree). A classical Oaxaca–Blinder decomposition focuses on average differences in wages and observable characteristics. The goal of this paper, however, is to analyse heterogeneity in the gender wage gap and to contribute to a more comprehensive understanding of gender inequality in income. This is particularly important to design efficient policies to establish equal pay for discriminated groups.

The extent to which the gender wage gap differs across women has attracted public attention and the interest of policy makers. While numerous policy reports and media articles have

¹ In line with the literature, we define the gender wage gap as the relative difference in earnings that emerges between female and male employees. To encode gender in our empirical analysis, we use the self-reported information provided in the ACS questionnaire item on a person's sex. More information available at <https://usa.ipums.org/usa/>

attempted to quantify heterogeneity in the wage gap, they have generally taken a simplistic approach based on comparing descriptive statistics across subgroups of people. In these studies, the subgroups are usually defined in terms of one characteristic only, such as region, age, race, ethnicity, or occupation (Baxter, 2015; Nelson, 2016; Overberg & Adamy, 2016; The American Association of University Women, 2018; U.S. Bureau of Labor Statistics, 2016; Vasel, 2017). Approaches such as this are likely to lead to flawed conclusions, however, because they neglect heterogeneity due to other observable variables.

We would like to shed light on potential drivers of the gender wage gap. Therefore, we suggest a model that allows for estimation of *ceteris paribus* changes of the gender wage gap according to observable characteristics—in other words, to estimate marginal effects keeping all other variables at fixed levels.

We recognise two major methodological challenges that are involved in a heterogeneity analysis of the gender wage gap. First, considering variation of the gender wage gap according to women's characteristics, which are very many, leads to a wage equation that includes a large number of explanatory variables. Hence, it becomes necessary to base estimation and statistical inference on modern machine learning methods and adjust for multiple testing and simultaneous inference. The second major challenge refers to quantification and visualisation of the gender wage gap according to women's characteristics. We provide quantile plots for the size of the gender pay gap together with uniformly valid confidence bands. To address these challenges, we base our analysis on a lasso estimator that is suitable to such a high-dimensional wage equation. To construct test statistics, *p*-values, and confidence intervals, we use the so-called post-double-selection approach by Belloni, Chernozhukov, and Hansen (2014), which provides a framework for valid inference based on the lasso estimator. In our analysis, we assess to what extent the gender wage gap varies according to various characteristics. This makes it necessary to appropriately account for multiple testing as otherwise, the probability of a type-I error, i.e. incorrectly rejecting at least one true null hypothesis, generally exceeds the nominal significance level α . We use recent statistical procedures to construct joint hypothesis tests and confidence bands for the double-selection estimator, which are attractive under considerations of statistical power.

In a first step of our analysis, we estimate heterogeneity in the coefficient of the female dummy variable in a Mincer Equation like wage regression. In a simple Mincerian wage equation, the gender coefficient (β) usually measures the average difference in wage between men and women holding all other covariates constant (*ceteris paribus*). We allow the coefficient to vary with socio-economic variables, x , giving rise to a regressor-dependent coefficient $\beta(x)$; in other words, the gender coefficient in the wage equation is determined for different subgroups. With the set of potential covariates being relatively large, we use recent machine learning methods, namely double lasso, for estimation of the effects and inference. In a second step, we illustrate the heterogeneity in the coefficients $\beta(x)$ in quantile plots and provide simultaneous confidence bands. The analysis reveals interesting patterns and shows that the gender wage gap is quite heterogeneous across different groups.

Our study contributes to the literature as it allows to estimate and quantify heterogeneity in the US gender wage gap in a data-driven way. Using data from the 2016 ACS, it provides an empirical assessment of a rich set of potential determinants of the gender pay gap apart from classical Oaxaca–Blinder decompositions. We aim to provide a more accurate picture of the gender wage gap than previous studies by allowing the gap to vary with a large number of individual characteristics. In doing so, we consider family and household-related demographic variables (i.e. marital status or having biological, adopted, or stepchildren at home), race, ethnicity (i.e. Hispanic origin), English language ability, geographic information (i.e. United States, census region and metropolitan statistical area), veteran status, labour market characteristics (i.e. industry, occupations and hours worked), and the classic human capital variables (i.e. potential labour market experience and years of education). For people with a bachelor's degree, we also include information on their college major. Considering a large number of explanatory variables creates a challenge in terms of statistical inference. To address this challenge, we employ up-to-date statistical methods that even allow for the number of variables to exceed the number of observations in the data set. In our analysis, statistical inference is performed using the lasso estimator in the double-selection approach by Belloni, Chernozhukov, and Hansen (2014). In the following, we will refer to this estimator as 'double lasso'. Moreover, we use recent methods to construct simultaneous confidence bands and joint hypothesis for the double lasso estimator which are attractive

under statistical power considerations. Furthermore, we contribute to the literature on gender inequality in earnings by estimating the gender wage gap for each woman in the data set and illustrating the resulting distribution of wage gaps for full-time employed women in the United States. We find that, in 2016, the US gender wage gap was highly heterogeneous and, thus, differed considerably from woman to woman depending on individual socio-economic characteristics. Whereas a substantial share of full-time employed women experienced a gap that exceeded the usually reported estimates, the estimated gender wage gap was non-significant for a considerable fraction of female full-time employees.

Summing up, we contribute to the literature in two regards: First, we analyse heterogeneity of the gender wage gap not only in one selected variable (as in most previous studies), but provide an extensive, data-driven analysis of the drivers of the gender wage gap in a high-dimensional setting with many potential drivers. Although establishing causal relationships is challenging in this setting, we provide a detailed analysis of potential drivers of the gender wage gap and give an extensive, descriptive analysis of the extent of the gender wage gap in various dimensions. These can form the basis of an in-depth analysis and for developing tailor-made policies trying to reduce the gender wage gap. We are not aware of such an extensive analysis in the previous literature. Second, methodology is presented for inference of heterogeneous treatment effects in high-dimensional settings, including quantile plots with uniformly valid confidence bands, building on recent theoretical contributions.

The paper is structured as follows. In Section 2, we give a short review on empirical methods in the context on the gender wage gap and recent developments in the literature. Section 3 presents the heterogeneous wage gap model together with the inferential framework. Section 4 introduces the data used in the empirical analysis, i.e. the ACS from 2016. In Section 5, we present results on heterogeneity of the US gender wage gap. In Section 6, we conclude and present an outlook for future research of the gender wage gap. Additional descriptive statistics and results are provided in the [online supplementary material](#).

2 Literature and evidence on the gender wage gap

2.1 Empirical methods in the context of the gender wage gap

Traditionally, decomposition methods as initially introduced in Oaxaca (1973) and Blinder (1973) are used in empirical studies that assess the gender wage gap. A detailed and comprehensive overview on decomposition methods and recent extensions thereof is provided in Fortin et al. (2011). The objective of the Oaxaca–Blinder decomposition is to distinguish whether the overall wage difference between men and women emerges due to gender differences in observable characteristics or due to a different valuation of these characteristics in the labour market, sometimes referred to as a ‘wage structure effect’ (Fortin et al., 2011). An example for the first effect is a situation with higher labour market experience, on average, for men than for women. Hence, if returns to labour market experience are positive, average earnings are higher for male employees than for female employees. The second effect emerges from the difference of the regression coefficients from two wage regressions that are separately estimated for male and female observations in the data. An example for the structural effect is a situation with higher returns to labour market experience for men than for women. In such a situation, women who have the same level of labour market experience earn on average less than men. Such a gender difference in valuations of labour market characteristics is often considered as an indicator of discrimination, although it might also reflect non-discriminatory effects, for instance unobserved productivity effects (Blau & Kahn, 2017). Recently, the econometric literature has developed innovative methodological extensions of the basic Oaxaca–Blinder decomposition that base upon quantile regression, for instance Chernozhukov, Fernández-Val, et al. (2013). These methods are able to detect heterogeneous patterns of the gender gap at different points of the income distribution. For instance, the gender wage gap was found to be more pronounced at the top of the income distribution than in the middle or at the bottom (Blau & Kahn, 2017).

Goldin (2014) provides a recent study of the gender wage gap that can be related to our approach. In the empirical analysis of Goldin (2014) that is based on ACS data, an ordinary least squares regression of an extended wage equation is estimated that included interactions of gender with a large number (i.e. 469) of occupation dummies. Being based on a theoretical argument, the

gender wage gap is allowed to vary across occupational categories, and, hence, the focus of the heterogeneity analysis is on variation by occupation. The results of Goldin (2014) illustrate the variation of the gender wage gap in an appealing way. Unfortunately, the significance of the effects is not reported. Under statistical considerations, however, the question of joint significance of heterogeneous effects is of great importance: If the number of tested hypotheses is large, adjustments for simultaneously testing multiple hypotheses are necessary in order to draw valid conclusions.

An approach that is related to our econometric framework has been recently developed by Chernozhukov, Fernández-Val, et al. (2018). Similar to the quantile plots, which we present in Section 5, the so-called sorted effects method provides estimates and confidence bands for an ordered sequence of partial effects that quantify heterogeneity in terms of observational characteristics. Indeed, the quantile plots in Section 5 coincide with the sorted effects if ordinary least squares regression is employed for estimation and if the same structural regression model is chosen in both approaches. Hence, our approach can be considered as a high-dimensional version of the sorted effects approach in Chernozhukov, Fernández-Val, et al. (2018) with the underlying estimation being based on the double lasso estimator. Our framework imposes a sparsity assumption, which says that only a subset of all variables have explanatory power for the outcome variable in the wage equation. Accordingly, the heterogeneity patterns in the gender wage gap are assumed to be sparse, too, i.e. only a small number of characteristics affect the variation of the gender wage gap. The sparsity assumption makes it possible to perform valid inference in a heterogeneity analysis in settings with many covariates, even if the number of explanatory variables exceeds the number of observations in the data (i.e. in high-dimensional settings). In these settings, classic estimation approaches such as ordinary least squares suffer from performance issues. We compare our results to those based on ordinary least squares regression in a robustness check. We find that in our setting, ordinary least squares results in noisy coefficient estimates and large standard errors and that more precise estimates can be obtained using the double lasso estimator. Another difference with regard to the sorted effects method refers to the bootstrap that is used to construct simultaneous confidence bands. In our approach, we use the multiplier bootstrap as developed in Chernozhukov et al. (2014) and Chernozhukov, Chetverikov, et al. (2013) to construct joint confidence regions, whereas (Chernozhukov, Fernández-Val, et al., 2018) rely on a classic bootstrap scheme. The multiplier bootstrap makes it possible to refrain from re-estimation of the coefficient based on bootstrap samples as it is instead based on permutations of an underlying score function, which makes it attractive under computational considerations. Yet another difference to the approach in Chernozhukov, Fernández-Val, et al. (2018) refers to the focus of the heterogeneity analysis. The so-called classification analysis in Chernozhukov, Fernández-Val, et al. (2018) provides an inferential framework for testing differences in observational characteristics of individuals in the most and least affected subgroup. In contrast, the focus in our study is on the variation of the gender wage gap estimate according to observational characteristics, in other words variation of $\beta(x)$ according to differences in x .

2.2 Literature review: the gender wage gap and recent developments

A great number of empirical studies have focussed on the gender wage gap, its determinants and its development over time and the life cycle. Due to the richness of the gender gap literature, we restrict attention to the literature on the gender gap in earnings and its determinants. Blau and Kahn (2017) provide an extensive and detailed review of various explanations of the gender wage gap together with an empirical reassessment of many theories.

The second half of the 20th century was characterised by a substantial convergence of the gender wage gap paralleled by a considerable convergence of men and women in terms of education, labour market experience and participation, and occupational choices, among others (Blau & Kahn, 2017; Goldin, 2014). A large part of the reduction of the gender wage gap that began in the 1980s and still continues until today, although in a less steady and slower manner, is attributed to the convergence in traditional human capital factors. Today, women achieve higher levels of education than men and almost the same levels of actual experience, on average. In a recent analysis, Blau and Kahn (2017) provide evidence that gender differences in observable characteristics such as experience, occupation and industry variables, explained two thirds of the total gender gap in 2010. As gender differences in terms of traditional human capital characteristics have

diminished over time, these factors have become less important in explaining the gender wage gap. For instance, in the decomposition of [Blau and Kahn \(2017, Table 4\)](#), differences in human capital characteristics could only explain 13% of the total gender wage gap in 2010 compared to 25% in 1980. Consequently, alternative explanations have been developed in the labour economics and sociological literature. In a recent analysis, [England et al. \(2020\)](#) assess to what extent gender-based differences in terms of various labour market characteristics persisted in recent decades. Their findings show that persistent differences in terms of labour market characteristics other than the classic human capital variables still contribute to the gender wage gap, for example segregation by field of study and occupation.

A recently proposed reasoning by [Goldin \(2014\)](#) focuses on the structure of jobs. Temporal flexibility, referring to factors like the total number of hours worked and the time when they are provided, translates into a convex relationship of working hours and the salary. Due to women's greater involvement in care work, e.g. in child rearing, gender inequality in earnings is expected to be more pronounced in inflexible occupations. [Goldin \(2014\)](#) presents evidence that the wage gap was larger and increased over the lifecycle in inflexible occupations, for example in the area of business or law, compared to more flexible occupations like pharmacy, science or technology. Moreover, in less flexible occupations, the gender wage gap was found to increase with the number of hours worked due to a more convex hours–earnings relationship. The explanatory power of occupations for the gender gap, together with industries, was also empirically confirmed in the analysis of [Blau and Kahn \(2017\)](#). [Cha and Weeden \(2014\)](#) and [Weeden et al. \(2016\)](#) provide a detailed analysis on the role of long work hours on the development of the gender wage gap over the last decades. They find that occupations with long work hours have been associated with a large growth in wages and hence contributed to the persistence and slow convergence in the gender wage gap observed in the 1990s and the 2000s as women, in particularly mothers, are less likely to work in these occupations.

The argumentation of [Goldin \(2014\)](#) and other studies is related to the fact that women are more likely to interrupt their work life because of having children and a greater responsibility in child rearing. Using data on actual labour market experience, [Blau and Kahn \(2017\)](#) emphasise the role of work life interruptions for the wage gap. In general, the effect of interruptions is relatively difficult to assess in empirical studies due to limited availability of actual labour market experience in many data sets. Moreover, explanations in favour of a ‘family’ or ‘motherhood penalty’ ([Pal & Waldfogel, 2016](#); [Sigle-Rushton & Waldfogel, 2007](#); [Waldfogel, 1998](#)) have been proposed and confirmed empirically implying that mothers tend to experience larger wage gaps than women without children. Recent studies, which mainly use administrative data from Scandinavian countries, assess the dynamics of the motherhood penalty over women's working history ([Albrecht et al., 2018](#); [Angelov et al., 2016](#); [Kleven et al., 2018](#)). A recently published study by [Bütikofer et al. \(2018\)](#) focuses on the motherhood penalty in high-paying jobs in Norway and assesses differences across four occupational categories. The analysis is based on the flexibility argumentation of [Goldin \(2014\)](#) and finds an association of greater motherhood penalties and occupations with lower flexibility. A detailed analysis of the motherhood penalty in the United States has recently been provided by [Pal and Waldfogel \(2016\)](#).

Furthermore, behavioural explanations suggest that psychological attributes and norms, for example weak preferences for competition and negotiations, cause gender differences in wages ([Manning & Swaffield, 2008](#); [Mueller & Plug, 2006](#)). However, [Blau and Kahn \(2017\)](#) conclude that these explanations cannot explain a large fraction of the gender wage gap and that further empirical non-labouratory evidence with stronger external validity is required to assess the importance of these theories.

Finally, taste-based or statistical discrimination is a potential source of the gender wage gap. The adjusted gender wage gap from an Oaxaca–Blinder decomposition is frequently taken as a measure of discrimination. However, the unexplained gap might as well be the result of unobserved factors related to productivity. Hence, there is no unambiguous empirical evidence of discrimination that is based on observational data. Real-world experiments point at a discrimination against women and mothers, for example ([Correll et al., 2007](#); [Neumark et al., 1996](#)). [Blau and Kahn \(2017\)](#) conclude that a part of the convergence of the wage gap in the 20th century might be explained by reduced discrimination against women in the labour market.

3 An econometric model of a heterogeneous gender wage gap

To motivate our approach, we start with a basic log wage regression where the coefficient β measures the relative difference in pay that emerges between men and women if one controls for the effects of observable characteristics, x_i . In the following, we use a variable on an individual's gender that is 1 if a person is female and 0 if male.

$$\ln w_i = \alpha + \beta \cdot \text{female}_i + x_i' \gamma + \varepsilon_i. \quad (3.1)$$

By construction of the wage equation, estimation of β in Equation (3.1) results in an average gender wage gap that is of the same magnitude for all women—even if they differ in terms of their observable characteristics. Hence, the resulting estimator will not be helpful in determining the driving forces of the wage gap nor to reveal heterogeneity of the wage gap. In order to model heterogeneity, we extend the basic wage equation in Equation (3.1) and let the gender coefficient $\beta = \beta(x_i)$ be a function of individual characteristics.

$$\ln w_i = \alpha + \beta(x_i) \cdot \text{female}_i + z_i' \delta + \varepsilon_i. \quad (3.2)$$

The $\beta(x_i)$ coefficient can be a linear or a more complicated function of the p_1 observable characteristics x_i , for example using transformations with splines or polynomials of higher order to approximate complex relationships of gender and the other explanatory variables. The covariates z_i in wage equation (3.2) are natural or constructed regressors, for instance it is possible to apply a so-called dictionary $p(x_i)$ to the initial regressors, x_i , to approximate the relationship of $\ln w_i$ and the observable characteristics. In our empirical application, we approximate $\beta(x_i)$ with a linear function of the regressors, i.e. $\beta(x_i) = \sum_{j=1}^{p_1} \beta_j \cdot x_{i,j}$ and the variables in z_i comprise all two-way interactions of the initial covariates x_i , including a constant. With this specification, the model corresponds to

$$\ln w_i = \alpha + \sum_{j=1}^{p_1} (\beta_j \cdot x_{i,j}) \cdot \text{female}_i + z_i' \delta + \varepsilon_i. \quad (3.3)$$

We consider p_1 initial characteristics x_i with corresponding coefficients β_j , $j = 1, \dots, p_1$, that enter $\beta(x_i)$. Together with the dimension p_2 of z_i and the corresponding vector of coefficients δ , the overall dimension of the model is $p = p_1 + p_2 + 1$. A negative β_j , $\beta_j < 0$, is interpreted as an increase of the absolute value of the wage gap. Hence, by default and in line with the presented empirical evidence in the literature, the 'gender wage gap' is interpreted as lower earnings for women, although the opposite might be observed in the data. More information on the interpretation is provided in [Section 1 of the online supplementary material](#) together with a short note on the relation to the Oaxaca–Blinder decomposition.

3.1 Valid post-selection inference in high dimensions

Studying the heterogeneity of the gender wage gap in the presented model requires to address two major challenges. First of all, including a large number of explanatory variables in Equation (3.3) makes it necessary to use modern statistical methods that can deal with high-dimensional data. We base our analysis on a lasso estimator that imposes a sparsity assumption. Hence, only a subset of all variables are assumed to drive the heterogeneity in the gender wage gap. The sparsity assumption makes it possible to overcome the performance issues that limit the use of ordinary least squares regression in settings with many explanatory variables. However, basing inference on machine learning methods such as the lasso prevents the direct use of standard inferential procedures such as a common t -test for a coefficient of interest. Instead, the inferential framework has to take the variable selection by lasso into account. We do so by employing the double-selection approach by [Belloni, Chernozhukov, and Hansen \(2014\)](#), which is introduced in the following paragraphs. The second major challenge in a heterogeneity analysis of the gender wage gap refers to simultaneously testing multiple coefficients. To model a heterogeneous gender wage gap, we include

multiple interaction terms of the female dummy with covariates x_i and summarise them in the coefficient $\beta(x_i)$. The corresponding test procedure needs to account for multiple testing as otherwise the probability of a type-I error (i.e. erroneously rejecting a true null hypothesis) exceeds the nominal significance level α . We apply recent methods to construct joint hypothesis tests and confidence bands for the coefficients in Equation (3.3) and $\beta(x_i)$, in particular we employ the multiplier bootstrap procedure by Chernozhukov et al. (2014) and Chernozhukov, Chetverikov, et al. (2013) in combination with the stepdown algorithm of Romano and Wolf (2005a), Romano and Wolf (2005b), and Romano and Wolf (2016).

The double-selection approach (Belloni, Chernozhukov, & Hansen, 2014) in combination with the work by Belloni, Chernozhukov, and Kato (2014) provides a uniformly valid inference framework for a vector of ‘target’ coefficients after model selection, for example as performed by the lasso. In our analysis, the target coefficients, which we want to test, refer to the interactions with gender, i.e. β_j , $j = 1, \dots, p_1$, in Equation (3.3). Hence, under a set of assumptions including sparsity, it is possible to perform valid post-selection inference even in cases where the number of regressors (p) exceeds the number of observations (n). In the context of the heterogeneous wage gap regression, basic lasso estimators $(\hat{\alpha}, \hat{\beta}(x_i), \hat{\delta})$ are defined as the solutions to

$$\begin{aligned} & (\hat{\alpha}, \hat{\beta}(x_i), \hat{\delta}) \\ & \in \arg \min_{\alpha, \beta(x_i), \delta} \left\{ \frac{1}{n} \sum_{i=1}^n (\ln w_i - (\alpha + \beta(x_i) \cdot \text{female}_i + z_i' \delta))^2 \right. \\ & \quad \left. + \frac{\lambda}{n} \|\Psi(\beta(x_i), \delta')'\|_1 \right\}, \end{aligned} \quad (3.4)$$

with Ψ being a diagonal matrix with data-dependent weights and $\beta(x_i)$ being specified as $\beta(x_i) = \sum_{j=1}^{p_1} \beta_j \cdot x_{i,j}$ in our empirical application. The lasso as initially developed in Tibshirani (1996) introduces a penalisation by the l_1 -norm of the coefficients to the least squares problem. This penalisation results in a shrinkage being applied to the regression coefficients towards zero. Finally, some of the coefficients are shrunk to a value exactly equal to zero such that the lasso performs a selection with respect to the set of regressors. Under the assumptions that sparsity holds in the underlying data generating process, solutions obtained with the lasso are sparse, in other words only relatively few, say s , of the p candidate regressors have explanatory power for the outcome variable. Sparsity avoids overfitting that is likely to arise in ordinary least squares regression with many regressors.

Estimation of Equation (3.4) requires a choice of the lasso penalty λ . Frequently, λ is set by (k -fold) cross-validation. However, since cross-validation is computationally expensive and the theoretical properties in high-dimensional settings are not yet fully developed, we determined λ by the theory-based rule of Belloni et al. (2012), which is also applicable to the case of heteroskedasticity. An intuitive introduction to the lasso and the reasoning of the penalty choice can be found in Belloni and Chernozhukov (2011). To be more exact, we estimated a post-selection version of the lasso, the so-called *post-lasso* (Belloni & Chernozhukov, 2013): The variables that have been selected by the lasso are used to set up a re-estimation step that is estimated by ordinary least squares regression, which produces the final coefficient estimates. The shrinkage performed by the lasso causes a bias of the coefficient estimates, which can be alleviated, in part, by using post-lasso.

Post-selection inference, in other words inference on regression coefficients after a model selection stage, has been an active research area in the statistics literature in the last years. In general, simply conducting ordinary least squares inference after estimation of Equation (3.4) with the lasso as if there was no variable selection does not result in valid statistical inference. Such an approach would only be feasible if one is willing to assume perfect model selection. The latter, however, is only guaranteed under a so-called *beta-min* assumption which requires that the non-zero coefficients of the true model are well distinguishable from zero.

The challenge for valid inference after a model selection step with the lasso or other machine learning methods is to avoid selection mistakes for variables that are both correlated with the

outcome and the target variables of interest, in other words incorrectly excluding *confounders* from the model. Removing a confounder from the regression model would introduce an omitted variable bias, which would effectively distort the estimate of the target coefficient. A valid approach to post-selection inference must therefore account for the risk of excluding confounders as illustrated in an intuitive example in [Belloni, Chernozhukov, and Hansen \(2014, Section 2.4\)](#). [Belloni, Chernozhukov, and Hansen \(2014\)](#) show that a naive selection procedure, which is based on a single selection step by lasso or a traditional t -test, can easily lead to an inconsistent estimator with a non-normal asymptotic distribution. In their example, they demonstrate the breakdown of the single-selection approach even in a setting where the effect of the confounding variable on the outcome is relatively close to zero.

The double-selection approach of [Belloni, Chernozhukov, and Hansen \(2014\)](#), and more generally, estimation based on orthogonalised moment conditions as in [Belloni, Chernozhukov, and Kato \(2014\)](#) offer an opportunity to overcome the problems of inference after a model selection stage. The idea of the method is to introduce an auxiliary lasso regression for every target coefficient to ensure that only moderate selection mistakes might occur. Double selection proceeds as follows:

1. For each of the p_1 target variables, estimate a lasso regression of the dependent variable in Equation (3.2), in w_i , on regressors z_i and the remaining targets. The target variables correspond to the interactions with gender in Equation (3.2).
2. Estimate an auxiliary lasso regression of each of the p_1 target variables on all remaining independent variables as regressors, i.e. the regressors z_i as well as the remaining targets.
3. Equation (3.2) is re-estimated with ordinary least squares regression with all variables being included that have been selected in either the first or the auxiliary regression steps.

The last estimation step, which includes all variables that have been selected by lasso either in the first or the auxiliary regression, reduces the risk that a confounding variable is excluded from the model. As a result, the double lasso produces valid standard errors and confidence intervals for the target parameters. Following [Belloni, Chernozhukov, and Hansen \(2014\)](#), [Belloni, Chernozhukov, and Kato \(2014\)](#), and [Chernozhukov, Hansen, et al. \(2018\)](#) and from asymptotic normality of the double-selection estimators, it is possible to show that under sparsity, $\hat{\beta}(x_i)$ as estimated by the double-selection approach asymptotically follows a normal distribution

$$\sqrt{n}(\hat{\beta}(x_i) - \beta_0(x_i)) \rightsquigarrow^d N(0, x_i' \Omega x_i), \quad (3.5)$$

with variance–covariance matrix Ω of the $\hat{\beta}_j, j = 1, \dots, p_1$, in $\hat{\beta}(x_i)$, which can be estimated according to [Belloni, Chernozhukov, and Hansen \(2014\)](#).

As the number of target parameters in the heterogeneous gender wage model is large, it is necessary to adjust for multiple testing. We implement the multiplier bootstrap procedure developed in [Chernozhukov et al. \(2014\)](#) and [Chernozhukov, Chetverikov, et al. \(2013\)](#) to construct uniformly valid confidence intervals for $\beta(x_i)$ and perform a valid joint test for the marginal effect targets $\beta_j, j = 1, \dots, p_1$, as suggested in [Belloni, Chernozhukov, and Kato \(2014\)](#). The multiplier bootstrap is based on an underlying orthogonal moment condition, which is perturbed using random weights. Based on this algorithm, it is possible to compute a constant c to construct joint confidence bands as well as p -values that are adjusted for multiple testing. The multiplier bootstrap procedure does not require that the coefficients $\beta_j, j = 1, \dots, p_1$, are re-estimated on bootstrap samples and, hence, is attractive under computational considerations. Moreover, we apply the stepdown procedure of [Romano and Wolf \(2005a\)](#), [Romano and Wolf \(2005b\)](#), and [Romano and Wolf \(2016\)](#) to adjust p -values in the joint hypothesis test, as recently established in [Chernozhukov, Chetverikov, et al. \(2013\)](#) and [Belloni, Chernozhukov, and Kato \(2014\)](#). The Romano–Wolf correction exploits the correlation structure that is underlying to the test statistics and proceeds in a stepwise manner, which makes it possible to increase statistical power as compared to traditional adjustments like the Bonferroni or Bonferroni–Holm procedure. For a more detailed presentation of the Romano–Wolf stepdown procedure and the underlying algorithm to construct p -values, we refer

to [Bach et al. \(2018\)](#). More technical details on the inferential framework for the regressor-dependent coefficient $\beta(x)$ is provided in the forthcoming work by [Chernozhukov, Hansen, et al. \(2018\)](#).

4 Heterogeneity of the US gender wage gap

4.1 Overview of the 2016 ACS data

In the empirical study, we use data from the 2016 ACS as provided by [Ruggles et al. \(2018\)](#) and extracted from the IPUMS-USA website.² The ACS provides a representative 1%-sample of the US population. Participation in the survey is mandatory. A large number of socio-economic characteristics at the individual and household level are available, for example referring to education, industry, and occupation. We restrict attention to employed individuals working full time (35+ hr) and year-round, i.e. at least 50 weeks a year, to compare men and women with a similarly strong attachment to the labour force. Weekly earnings are computed as annual earnings divided by 52 (weeks). We focus on individuals aged 25–65 and discard persons with income below the mandated federal minimum level of wages corresponding to an hourly wage of 7.25 or—in terms of annual wage income—to 12,687.50 according to our sample composition. As the federal minimum level has not been adjusted since 2009, we consider our exclusion rule as not restrictive. However, the rule is sufficient to exclude unrealistic weekly wages, for instance wages corresponding to less than 1 per hour. The final data set comprises 642,229 individual observations and is stratified into two subgroups according to individuals' highest educational degree. The 'bachelor's degree data' comprises 288,095 individuals with at least a bachelor's degree and the 'high school degree data' consists of 354,134 observations with at most graduation from high school, GED or equivalent.

4.2 Descriptive statistics

[Table 1](#) provides summary statistics for a selection of variables available in the 2016 ACS data. The descriptive statistics illustrate that wages are substantially higher for college graduates on average. As expected, the individuals holding at least a bachelor degree are in education for a longer time and have less potential labour market experience, on average. The shares of Hispanics and Blacks are lower in the bachelor's degree subgroup, whereas the share of Chinese is higher. College graduates tend to live in metropolitan statistical areas more frequently and to work longer hours, on average. Also the shares of persons who live with their biological, adopted or stepchildren aged 18 or younger are higher in the bachelor's degree data. Similarly, the share of persons who reside with their biological, adopted or stepchildren aged four or younger is higher in the sample of college graduates. The patterns in terms of marital status differ across the educational attainment subgroups. The share of married (with spouse present) persons is higher in the bachelor's degree data.

In both samples, average earnings of men exceed those of women by far, both in terms of the mean (around 32% for the high school and 49% for the bachelor's degree sample) and the median weekly wage (33% and 42%). The data set reflects a common observation with regard to the human capital characteristics years of education and potential labour market experience. The summary statistics for the high school degree data reflect the frequently mentioned reversal of the gender gap in terms of labour market characteristics ([Blau & Kahn, 2017](#)). However, we cannot confirm this observation for the sample of college graduates, probably due to selection into full-time employment. The gender gap in terms of years of education is virtually zero. Moreover, we observe that the gender gap in terms of hours worked is still considerable with men working for about 2.4 (bachelor's degree) and 3 (high school) hours each week longer than their female counterparts, on average. [Figure 1](#) illustrates the fact that the share of men in the group of employees who regularly work overtime is disproportionately large. [Figures S1–S6 in the online supplementary material](#) provide further insights on the distribution of the observable characteristics given gender and educational attainment.

The abovementioned decrease of the gender wage gap in terms of human capital characteristics in the high school degree subgroup are in concordance with the estimated female-to-male wage ratios shown in [Figure 2](#). We base the log wage ratio analysis on that in [Blau and Kahn \(2017\)](#),

² <https://usa.ipums.org/usa/>

Table 1. Summary statistics

Variable	High school degree data		Bachelor's degree data	
	Men	Women	Men	Women
Weekly wage (mean)	1,098.95 (863.34)	833.83 (618.87)	2,244.29 (1,996.49)	1,508.86 (1,240.98)
Weekly wage (median)	923.08	692.31	1692.31	1192.31
Single/never married	0.21 (0.16)	0.19 (0.16)	0.19 (0.15)	0.24 (0.18)
Married, spouse present	0.63 (0.23)	0.54 (0.25)	0.72 (0.20)	0.60 (0.24)
Child age ≤ 4	0.12 (0.11)	0.08 (0.07)	0.16 (0.14)	0.13 (0.11)
Child age ≤ 18	0.37 (0.23)	0.33 (0.22)	0.44 (0.25)	0.38 (0.24)
White	0.86 (0.12)	0.81 (0.16)	0.84 (0.14)	0.81 (0.16)
Black	0.10 (0.09)	0.15 (0.12)	0.05 (0.05)	0.09 (0.08)
Chinese	0.01 (0.01)	0.01 (0.01)	0.03 (0.03)	0.03 (0.03)
Hispanic	0.14 (0.12)	0.12 (0.11)	0.06 (0.05)	0.06 (0.06)
Potential experience (years)	27.03 (11.23)	28.48 (11.08)	21.61 (10.99)	20.38 (11.19)
Years of education	12.43 (1.17)	12.66 (1.06)	16.95 (1.28)	16.97 (1.23)
Hours worked (mean)	44.93 (8.56)	41.90 (6.34)	46.20 (8.65)	43.79 (7.34)
Hours worked (median)	40	40	43	40
Veteran status	0.11 (0.31)	0.01 (0.12)	0.07 (0.25)	0.02 (0.12)
MSA	0.85 (0.36)	0.86 (0.35)	0.94 (0.23)	0.93 (0.26)
No. of observations (%)	207,549 58.61	146,585 41.39	154,833 53.74	133,262 46.26

Note. Mean and median values for selected observable characteristics. Standard deviation in parantheses. *Data:* Subsample of the ACS 2016 according to sample composition explained in the text. MSA = metropolitan statistical area.

Chapter 2.1). Accordingly, the female-to-male wage ratio is slightly smaller if we conditioned on human capital factors. The resulting wage ratios are 77% if we control for human capital characteristics and 78% if we consider the unconditional gap. If we condition on additional individual characteristics including occupation, industry and hours worked, among others, average wages of female employees are around 17% lower than wages of the male employees.³ The patterns observed for the sample of academics reveal that conditioning on human capital factors lifts the

³ The 17% wage loss corresponds to the 'average female residual from the male wage equation' in [Blau and Kahn \(2017, p. 800\)](#).

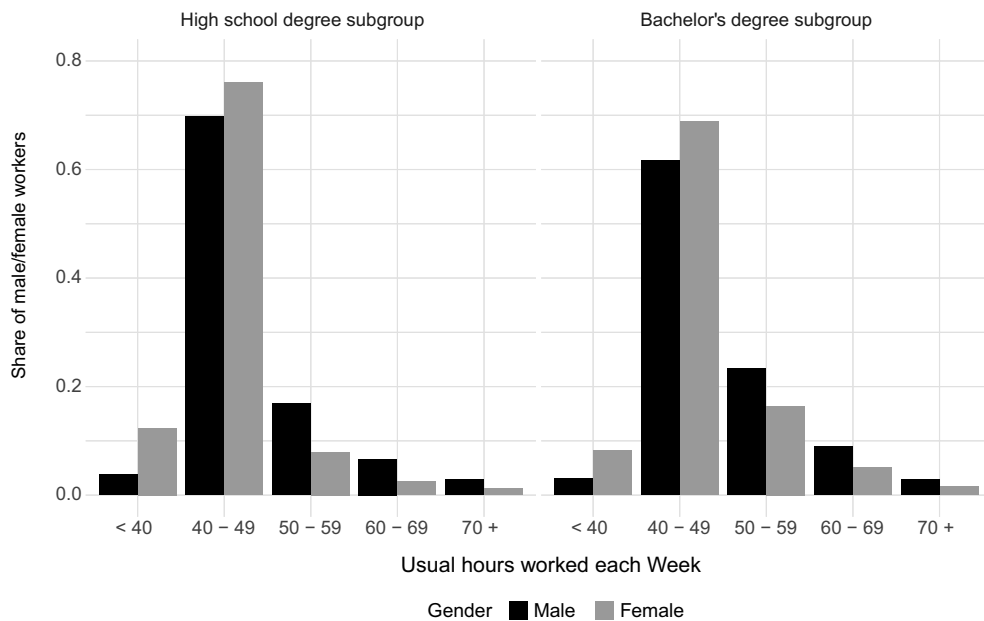


Figure 1. Usual hours worked per week by gender. The bar plot in the figure illustrates the distribution of male (black bars) and female (grey bars) employees across categories of usual hours worked each week, separately for the two educational attainment subgroups.

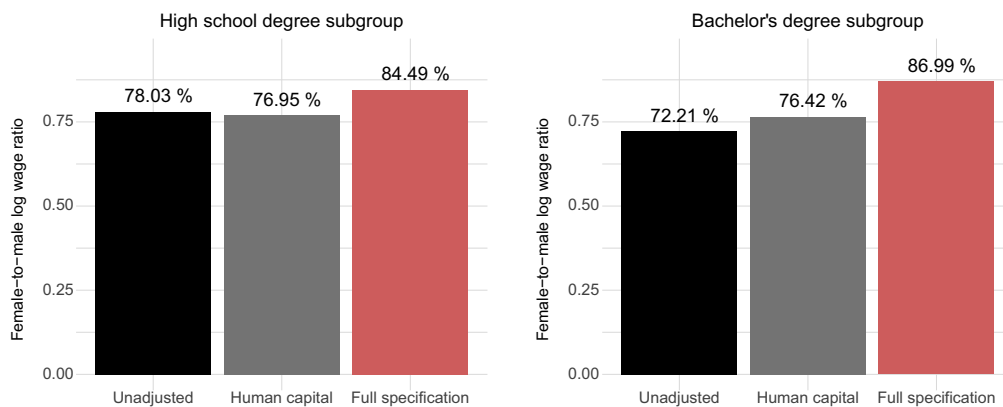


Figure 2. Unconditional and conditional female-to-male mean (log) wage ratios. The bar plots indicate the female-to-male log wage ratios, i.e. the quotient $\exp(\bar{X}_f \gamma_f) / \exp(\bar{X}_f \gamma_m)$, with regression coefficients γ_f and γ_m from a regression separately performed for the female and male observations. The unadjusted wage ratio corresponds to $\exp(\log(w_f)) / \exp(\log(w_m))$. The human capital specification includes the regressors years of education, potential experience, and potential experience squared. In the full regression specification, variables on industry, occupation, the number of hours worked each week and, for the bachelor's degree data only, the field of college major are included. In the regressions, we additionally control for English language ability; veteran status; US census region; race; Hispanic origin; binary variables indicating if a person lives in the same household with a biological, adopted or stepchild children of age 4 or younger and, respectively, of age 18 or younger; metropolitan statistical area; and marital status.

female-to-male wage ratio from a level of 72%–76%. Including additional individual characteristics lead to an estimated wage ratio of 87% corresponding to a residual wage gap of approximately 14%.

In the empirical analysis, we use a set of 16 initial regressors to model heterogeneity in the gender wage gap. The variables are listed in Table 2 together with information on the baseline

Table 2. List of variables

Variable	Type	Baseline category
<i>Dependent variable</i>		
Log weekly wage	Continuous	
<i>Independent variables</i>		
Female	Binary	
Marital status	6 categories	Never married/single
Child age ≤ 4 (one or more biological, adopted, or stepchildren at home aged 4 or younger)	Binary	
Child age ≤ 18 (one or more biological, adopted, or stepchildren at home aged 18 or younger)	Binary	
Race	4 categories	White
Hispanic	Binary	
English language ability	5 categories	Speaks only English
Potential experience (years)	Continuous	
Potential experience squared	Continuous	
Years of education	Continuous	
Veteran status	Binary	
Industry	14 categories	Wholesale trade
Occupation	26 categories	Management, business, science, and, arts
Hours worked each week (usually worked in last 12 months)	5 Categories	35–40 hr
College major (bachelor’s degree data only)	37 categories	Education administration and teaching
Region (US census)	9 categories	New England division
MSA (metropolitan statistical area)	Binary	

categories. The dependent variable in the wage regression is log weekly wage implying that wage gap estimates are reported in log scale throughout the paper. We model parenthood by including two binary variables. The first of these variables indicates that a person resides with one or more biological, adopted or stepchildren of age 18 or younger. The second variable takes on value one if a person lives in the same household with a biological, adopted or stepchild aged 4 or younger. We include both variables to analyse heterogeneity in the motherhood penalty in terms of the age of the child. We use the 14 major groups of the 1990 Census Bureau industry classification scheme available in the ACS (3-digit). Similarly, the Census Bureau provides a 2010 ACS classification of occupations (4-digit) that are clustered into 26 major categories in the ACS. The variable on hours worked is a categorical variable indicating the number of hours usually worked each week in the last 12 months. For the bachelor’s degree subgroup, we additionally include the variable college major to account for individuals’ educational background in more detail. The exact coding of the categories and definition of the variables can be found on the corresponding documentation website of IPUMs.

To model heterogeneity in the wage gap, we construct all two-way interactions of the initial regressors and end up with a high-dimensional setting with a value of p that corresponds in total to 2,068 (high school degree subgroup) and 4,382 (bachelor’s degree subgroup) regressors

(categorical variables are transformed to level-wise dummies and variables with zero-variation are dropped). Of these regressors, 71 (high school) and 106 (bachelor) refer to the initial set of characteristics x_i and dimensions p_1 and 1,997 (high school) and 4,276 (bachelor) to the interacted regressors z_i and dimension p_2 in regression equation (3.2) in Section 3.

In the following, we will selectively report the main findings. More detailed results are available in the [online supplementary material](#). For example, [online supplementary Tables S1–S7](#) present the coefficient estimates and p -values for all coefficients. Moreover, a graphical illustration of the joint confidence bands is presented in [Figures S7 and S8 of the online supplementary material](#).

5 Results

5.1 Significant variables for the US gender wage gap

In a first step, we focus on the most important drivers of the gap emerging between men's and women's weekly wages. [Tables 3 and 4](#) present selected β_j -estimates from Equation (3.3) that are jointly significant at the 5%-level. The p -values are obtained from a joint significance test using the multiplier bootstrap procedure developed in [Chernozhukov et al. \(2014\)](#) and [Chernozhukov, Chetverikov, et al. \(2013\)](#) and suggested for valid simultaneous inference after model selection in [Belloni, Chernozhukov, and Kato \(2014\)](#) in combination with the stepdown procedure of [Romano and Wolf \(2005a\)](#) as recently established in [Chernozhukov, Chetverikov, et al. \(2013\)](#) and [Belloni, Chernozhukov, and Kato \(2014\)](#). The estimated coefficients of discrete regressors indicate changes of the gender wage gap as compared to the gap in the baseline group.⁴ For instance, the results obtained for the high school degree subgroup show that married women (spouse present) experience a wage gap that is *ceteris paribus* about 10–11 percentage points (pp) larger (in absolute terms) than that of never married women, on average.

Overall, we find that the gender gap varied substantially with individual characteristics in both educational attainment subgroups. The gender wage gap is found to change heavily with family and household-related characteristics (i.e. marriage and motherhood), race, and labour market conditions (i.e. industries and occupations). Patterns with respect to the organisation of the household and family as well as race exhibit similarities in the two subsamples. The effects associated with job-related variables like industry, occupation, and hours worked differ in sign and magnitude across the two educational attainment subgroups.

5.1.1 Family and household-related characteristics

We find that, in 2016, the gender wage gap of married women is *ceteris paribus* around 9–11pp larger than that of women who have never been married. The magnitude of the marriage effect is relatively large also if compared to the motherhood penalty. According to the results, mothers, who are defined as women living in the same household with at least one biological, adopted or stepchildren aged 18 or younger, have a gender gap that is approximately 5–6pp larger than that of women who do not reside with a child aged 18 or younger. In both samples, the wage gap for mothers of young children—defined as women who live in the same household with at least one biological, adopted, or stepchild aged 4 or younger—is smaller than for those with older children. This effect is particularly sizeable in the bachelor's degree subgroup.

These women experience an overall reduction of the wage gap by more than 2pp as compared to women who do not reside with a biological, adopted or stepchild (of age 18 or younger). These patterns on the motherhood penalty are in line with previous findings in [Pal and Waldfogel \(2016\)](#) who report a larger wage gap for mothers of older children as compared to mothers with the youngest children being younger than age 6. We interpret the results as evidence of a time-persistent motherhood penalty, at least for mothers with a strong labour force attachment and high levels of education, rather than an immediate wage reduction due to child birth. As can be seen from the descriptive analysis in Section 4.2, men are more likely to work full-time than women in both subsamples. Hence, the results on motherhood are likely to be affected by selection into full-time employment and we therefore interpret our findings rather as descriptive evidence, which deserves further investigation in future research.

⁴ The gender wage gap in the baseline group is indicated by the constant β_0 , i.e. the average gender gap experienced by women with characteristics never married, White, wholesale trade industry, ...All baseline definitions are listed in [Table 2](#).

Table 3. Double lasso results

Variable	High school degree subgroup		Bachelor's degree subgroup	
	Estimate	p-Value	Estimate	p-Value
Constant	−0.0463	0.9070	0.0428	1.0000
<i>Marital status</i>				
Married, spouse pres.	−0.1096	0.0000	−0.0973	0.0000
Married, spouse abs.	−0.0737	0.0010	−0.0535	0.2630
Separated	−0.0575	0.0030	−0.1205	0.0000
Divorced	−0.0571	0.0000	−0.0548	0.0000
Widowed	−0.0536	0.0700	−0.1152	0.0110
<i>Child</i>				
Age 18 or younger	−0.0507	0.0000	−0.0531	0.0000
Age 4 or younger	0.0289	0.0180	0.0809	0.0000
<i>Race</i>				
Black/African American	0.0789	0.0000	0.0679	0.0000
Chinese	0.0819	0.0100	0.0589	0.0020
Other Asian or Pacific Isl.	0.0716	0.0000	0.0437	0.0010
<i>Veteran status</i>				
Veteran	0.0429	0.0140	0.0204	0.9930
<i>Potential experience</i>				
Exp	−0.0040	0.0000	−0.0024	0.2770
Exp2	0.0000	0.3180	−0.0000	0.9270
<i>Years of education</i>				
Years of schooling	−0.0026	0.920	0.0056	0.1560
<i>Industry</i>				
TRANS	−0.0535	0.0030	0.0217	1.0000
RETAIL	−0.0444	0.0150	−0.0216	1.0000
FINANCE	−0.0493	0.0180	−0.0799	0.0000
BUISREPSERV	−0.0433	0.0640	−0.0557	0.0450
PROFE	−0.0742	0.0000	−0.0668	0.0000
ADMIN	−0.0527	0.0140	−0.0091	1.0000
<i>Usual hours worked</i>				
40 to 49	−0.0456	0.0000	−0.0104	1.0000
50 to 59	−0.0374	0.0150	−0.0048	1.0000
60 to 69	−0.0534	0.0150	−0.0207	0.9980
>70	−0.1186	0.0000	−0.0623	0.2150
<i>College major</i>				
Comp/Inform Sci	−	−	−0.0666	0.0000
Engin	−	−	−0.0545	0.0000
Bio/Life Sci	−	−	−0.0496	0.0040
Math/Stats	−	−	−0.0683	0.0110
Phys Sci	−	−	−0.0570	0.0040
Psych	−	−	−0.0705	0.0000
Crim Just/Fire Prot	−	−	−0.0788	0.0000

(continued)

Table 3. Continued

Variable	High school degree subgroup		Bachelor's degree subgroup	
	Estimate	<i>p</i> -Value	Estimate	<i>p</i> -Value
Soc Sci	–	–	–0.0613	0.0000
Bus	–	–	–0.0621	0.0000
Hist	–	–	–0.0561	0.0440

Note. Selected results from post-lasso estimation using double selection (double lasso). Coefficients printed in black are significant at the 5% level. *p*-Values are obtained from a joint test of all β_j coefficients in $\beta(x_i)$ from Equation (3.2) using the multiplier bootstrap procedure suggested in Belloni, Chernozhukov, and Kato (2014) with 1,000 repetitions in combination with the stepdown procedure of Romano and Wolf (2005a). Results on occupation are presented in Table 4.

5.1.2 Race and ethnicity

The results suggest that there is substantial heterogeneity of the gender gap according to race. Women in race categories other than White experience a significantly smaller gender wage gap. The race-based gender gap differentials are more pronounced in the group with lower educational attainment where the wage gap for non-Whites is 7–8pp smaller than for Whites. Hence, the frequently reported variation of the gender gap according to race is robust to controlling for a large set of characteristics such as years of education, potential experience, occupation, industry, and level of English language ability, even in the bachelor's degree subgroup. Controlling for observable characteristics render the gender gap differential non-significant only for individuals in the group of Hispanics.

5.1.3 Education and potential experience

As we will point out in Section 5.2, the heterogeneity patterns of the gender gap for full-time and year-round employed women vary substantially across the two subgroups. Within the two samples, however, the gender wage gap is found to only vary moderately according the classic human capital variables such as years of education and potential labour market experience. Splitting the sample into two subgroups according to individuals' highest degree seems to capture a large proportion of the variation of the wage gap in terms of educational attainment. The coefficient on the interaction of the female variable with years of education is not significant in both samples. The effect of potential labour market experience is smaller than 0.01 and significant only for the high school degree subgroup, indicating that for this group the gender gap increases gradually over the employment history. The non-significant coefficient on experience squared points at a weak linear relationship of potential labour market experience and the magnitude of the wage gap. Whereas the coefficient on years of education is non-significant, the gender gap of individuals with postsecondary education is found to vary by several college categories. We found that the gender gap is significantly larger than that in the baseline category 'Education Administration and Teaching' in 10 out of the 36 college majors, among others, in natural science disciplines such as 'Biology and Life Science', 'Physical Sciences', as well as in the categories 'Social Sciences' and 'Business'. In the wage regression, we additionally include the level of English language ability and do not find a significant effect on the magnitude of the gender gap.

5.1.4 Industries, occupation, and hours worked

Previous studies provided evidence that the gender wage gap varied according to job-related characteristics associated with specific industries and occupations (Blau & Kahn, 2017; Goldin, 2014).

We find distinct patterns of job-related effects in both educational attainment subgroups. Our results on the number of hours usually worked each week provide evidence that the gender wage gap for women with at most a high school degree is around 3.7–5.3pp larger for those in the categories with 40–69 hr as compared to women in the baseline category (35–40 hr). The gender wage gap for women working more than 70 hr per week is found to be around 12pp larger than that in the baseline group. The coefficient estimates on long work hours are smaller and non-significant for the sample of college graduates.

Variation by industry is found to be more pronounced for the high school degree subgroup. For this subgroup the variation of the gap is significant in five industries as compared to only three significant coefficients observed for persons with a college degree. The patterns for the industries ‘Finance, Insurance, and Real Estate’ and ‘Professional and related Services’ are similar in both subgroups corresponding to a larger wage gap by 5–8pp than for the baseline industry. For people with at most a high school degree, we observe larger wage gaps in the industries ‘Transportation, Communications, and other Public Utilities’, ‘Retail Trade’, and ‘Public Administration’ compared to the baseline category ‘Wholesale Trade’. Heterogeneity of the wage gap by occupation tends to be more pronounced than heterogeneity by industry which is generally in line with the argumentation and results in Goldin (2014). The magnitude of occupational effects is larger than that of industry effects and the patterns tend to be different in the two educational attainment subgroups. Table 4 presents the estimates of the occupational effects in an increasing order. Apparently, the difference in wages between male and female employees are particularly small in occupations in the category ‘Community and Social Services’ (approximately 12pp smaller gender gap for the high school degree subgroup and 17pp for the bachelor’s degree data), as well as in ‘Healthcare Practitioners and Technical’, ‘Construction’ and ‘Installation, Maintenance, and Repair’ for persons with at most a high school degree. The patterns observed for college graduates are different. Jobs in the category ‘Healthcare Practitioners and Technical’ are associated with a relatively large gender wage gap. Highly educated employees in ‘Education, Training, and Library’ occupations experience a small wage gap as compared to other occupations. In contrast, this occupational category is associated with the largest occupational effect on the gender wage gap for the high school degree subgroup. Moreover, the wage gap for employees with a bachelor’s degree working in a scientific or technical occupation (e.g. ‘Architecture and Engineering’ or ‘Life, Physical, and Social Science’) experience a wage gap that is relatively small as compared to the baseline category.

In general, the different patterns of job-related effects in the two subsamples can be justified with an extension of the flexibility-based argumentation of Goldin (2014). Basically, the effect of higher education on flexibility can go either way: Higher educational achievement might be associated either with an increase or a decrease in job flexibility. On the one hand, higher education, in other words, the acquisition of more abstract skills, might make it easier for women to work in a more flexible work environment given an occupational category, for example in education or science. Whereas on the other hand, higher education might allow to enter jobs, that are in general less flexible than jobs for less-educated in the same occupational category. For example, compared to a nurse, a surgeon as a healthcare practitioner might be less able to work fewer hours or decide when to provide these hours.

5.2 Heterogeneity in the coefficient function $\beta(x_i)$

Recognising heterogeneity in the gender wage gap is highly relevant for policy makers as it might help them to choose efficient and effective policy measures. In order to contribute to a more comprehensive understanding of gender inequality in earnings, we estimate the gender gap for every woman in the data set. The $\beta(x_i)$ coefficient from Equation (3.2) summarises all effects attributable to the observed characteristics, and hence, gives an estimate on the gender wage gap of a woman with characteristics x_i . Moreover, the inferential framework presented in Section 3.1 makes it possible to construct confidence bands for $\beta(x_i)$. Thus, we are able to judge whether a women’s wage gap is meaningful from a statistical point of view. We present quantile plots of the individual gender gap estimates together with confidence bands as obtained for all women in the two educational attainment subgroups in Figure 3. We additionally report the ordinary least squares results to allow for comparison.

The quantile plot illustrates that the US gender wage gap for full-time and year-round employed women was highly heterogeneous in 2016. Rather than affecting all women to the same extent, gender inequality in wages consists of a range of wage penalties that differ greatly from woman to woman. For most women, the estimated gap deviates from the abovementioned estimates, derived from traditional analysis. In Section 4.2, we reported an unadjusted gap of approximately 24% (high school degree subgroup) and 33% (bachelor’s degree subgroup) and adjusted wage gaps of 17% and 14%, respectively.

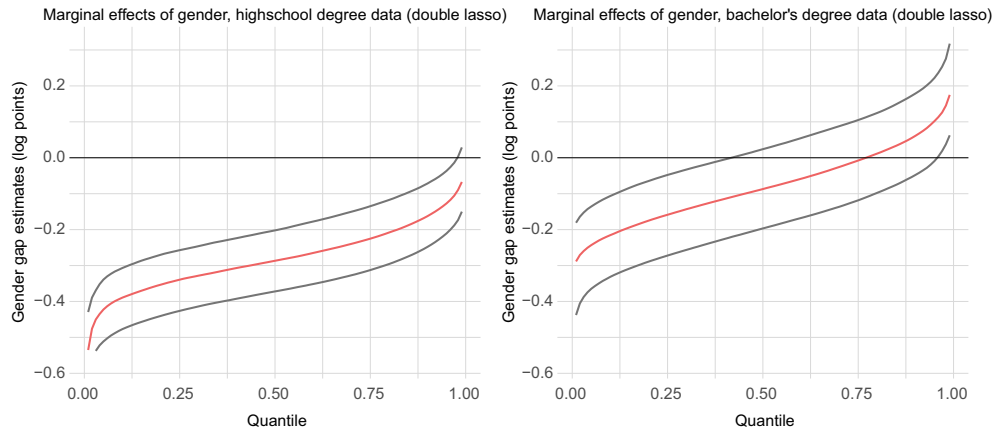
Table 4. Ordered occupational effects, high school degree and bachelor's degree data, double lasso

High school degree subgroup			Bachelor's degree subgroup		
Variable	Coefficient	<i>p</i> -Value	Variable	Coefficient	<i>p</i> -Value
Educ/Training/Libr	−0.1836	0.0000	Healthc Supp	−0.1022	0.3950
Extract	−0.1448	0.9600	Milit Specific	−0.0799	0.9930
Prod	−0.0974	0.0000	Farm/Fish/Forestry	−0.0498	1.0000
Arts/Design/Entert/	−0.0304	0.9200	Office/Administr Supp	−0.0465	0.0000
Sports/Media			Healthc Pract/Technic	−0.0407	0.0260
Sales	−0.0187	0.7210	Financ Spec	−0.0348	0.0690
Financ Spec	−0.0127	0.9760	Pers Care/Serv	−0.0287	1.0000
Build/Grounds Clean/	−0.0108	0.9710	Build/Grounds Clean/	−0.0248	1.0000
Mainten			Mainten		
Transp	−0.0085	0.9760	Sales	−0.0162	0.9980
			Food Prepar/Serving	−0.0011	1.0000
Archit/Engin	0.0189	0.9760	Prod	0.0065	1.0000
Pers Care/Serv	0.0200	0.9200	Transp	0.0228	1.0000
Comput/Math	0.0246	0.7830	Comput/Math	0.0372	0.0020
Farm/Fish/Forestry	0.0286	0.9760	Bus Operat Spec	0.0377	0.0110
Food Prepar/Serving	0.0290	0.2360	Arts/Design/Entert/	0.0469	0.0330
Milit Specific	0.0391	0.9760	Sports/Media		
Technic	0.0419	0.9200	Legal	0.0495	0.0810
Protect Serv	0.0479	0.0510	Educ/Training/Libr	0.0606	0.0000
Healthc Supp	0.0530	0.0420	Archit/Engin	0.0620	0.0000
Bus Operat Spec	0.0571	0.0030	Protect Serv	0.0666	0.0110
Office/Administr Supp	0.0635	0.0000	Life/Physical/Soc Sci.	0.0719	0.0000
Life/Physical/Soc Sci.	0.0703	0.5360	Technic	0.1126	0.5270
Install/Mainten/Rep	0.0742	0.0070	Constr	0.1469	0.0810
Constr	0.0895	0.0150	Install/Mainten/Rep	0.1496	0.0020
Legal	0.1042	0.3400	Comm/Soc Serv	0.1702	0.0000
Healthc Pract/Technic	0.1075	0.0000			
Comm/Soc Serv	0.1205	0.0000			

Note. The table presents occupational effects for the high school degree subgroup (left) and the bachelor's degree subgroup (right) in increasing order to provide a comparison of the occupational patterns observed for both subgroups. *p*-Values are obtained from a joint test of all β_j coefficients in $\beta(x_i)$ from Equation (3.2) using the multiplier bootstrap procedure suggested in Belloni, Chernozhukov, and Kato (2014) with 1,000 repetitions in combination with the stepdown procedure of Romano and Wolf (2005a). Significant (printed black) and non-significant (printed grey) coefficients at a 5% significance levels are presented.

Patterns of heterogeneity varied substantially across the two samples, with gender wage inequality being more prevalent and more severe among women with lower educational attainment. Whereas more than 90% of female employees with a highschool degree or lower earn significantly less than their male counterparts, this number is much smaller for those females who hold at least a bachelor's degree—according to the double lasso results in Panel (a) of Figure 3. Moreover, at any given quantile, the wage gap is larger for women who do not have a college degree. The median of the estimated wage gaps is around 9% (non-significant) for women with postsecondary education. In contrast, half of the women with at most a high school degree experience a wage gap of at least 29%. Interestingly, there is evidence of a reversal of the gender wage gap for a small share of women with a college degree, i.e. 4% of the full-time and full-year employees with postsecondary education earn significantly more than comparable men according to our double lasso results.

(a) Quantiles of effects with corresponding confidence bounds, DOUBLE LASSO.



(b) Quantiles of effects with corresponding confidence bounds, OLS.

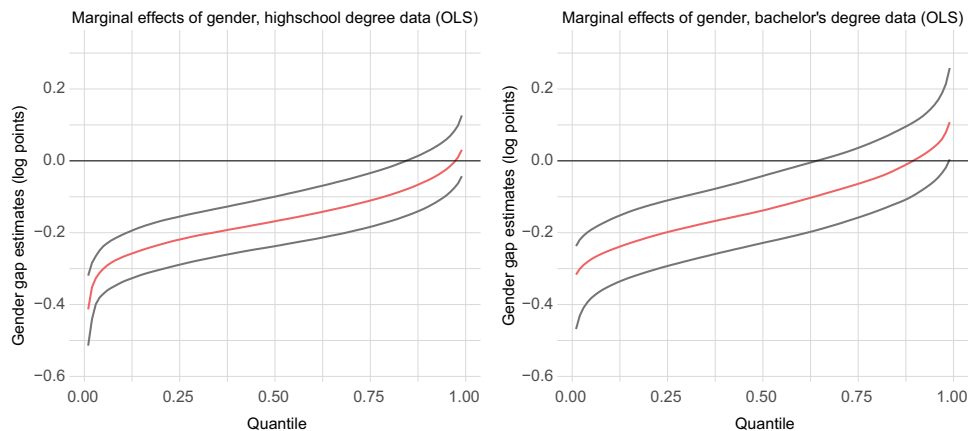


Figure 3. Quantiles of effects with simultaneous confidence bands. The plots show the quantiles of the individual gender wage gap estimates as computed for all women in the educational attainment subgroups of the ACS 2016 data together with simultaneous 0.95 confidence bands (grey lines) obtained from the multiplier bootstrap procedure with 500 repetitions. Estimates in Panel (a) are obtained from a high-dimensional wage regression using the double lasso estimator, with log weekly wages as the dependent variable. Plots on the left refer to the high school degree subgroup and plots on the right to the bachelor's degree subgroup. In addition, ordinary least squares results are provided in Panel (b) for reasons of comparison.

5.3 A robustness check

We perform a robustness check with regard to the degree of penalisation in the lasso estimation steps. The lasso estimator that is based on a theoretical choice of the penalty term λ involves a constant c_{penalty} . A lower value of this parameter is associated with a less severe penalisation. The corresponding quantile plots and result tables are presented in the [online supplementary material](#). The major conclusions drawn in the previous section continue to hold in the setting with less severe penalisation. Moreover, we compare the results in Panel (a) of [Figure 3](#) to ordinary least squares estimation presented in Panel (b) of [Figure 3](#). However, we observed difficulties of the ordinary least squares estimator in estimation of the covariance matrix. The covariance matrix is used in the multiplier bootstrap procedure to construct the simultaneous confidence bands. Hence, the corresponding results must be interpreted with caution.

6 Conclusion

We started with the objective to analyse heterogeneity in the gender wage gap. Our attempt to answer this question provided detailed insights to the determinants of the gender wage gap and

their consequences in the aggregate. Thereby we considered characteristics related to the organisation of the family and household, race and ethnicity as well as job-related information. Compared to previous studies we did not analyse heterogeneity with regard to one pre-specified variable, but determined the drivers of the heterogeneity in a data-driven way among a large set of potential variables employing modern machine learning methods.

Our analysis was built on recent approaches to valid statistical inference that allow for the use of machine learning methods. By using the double lasso estimator, which exploits a sparsity assumption, we were able to improve estimation precision as compared to a framework that is based on classic ordinary least squares. We noticed that double lasso did not exhibit the instabilities of least squares regression with respect to estimation of the covariance matrix, which in turn allowed for less conservative results for simultaneous inference. Although our primary results closely align with both the least squares analysis and a robustness check utilising less severe lasso penalisation, our model offers novel insights and contributes significantly to the literature on the gender wage gap. In complex and high-dimensional settings, particularly those with non-linear relationships, we anticipate that our machine learning-based approach will outperform traditional methods even more.

In summary, our empirical analysis reveals that in 2016, most full-time employed women in the United States experienced a substantial wage penalty compared to otherwise identical men. However, the extent to which women were affected by gender inequality in earnings differed greatly according to individual characteristics, including educational attainment, marital status, having children at home, race, and job-related characteristics such as occupation and industry. The commonly used average estimates of the gender wage gap can therefore be seen as a poor approximation of the wage penalty that is experienced by most women. By illustrating and quantifying heterogeneity in the wage gap, we hope to contribute to both the public and the academic discussion, and to provide information that policy makers can use to design more effective policies. In the following, we attempt to briefly outline a proposal for policy makers that takes our results into account. Our study serves to generate new directions for policy measures to reduce gender inequality but of course these need to be validated with further empirical analyses. Future policies might focus on improving work conditions for women without a college degree as we find that this group is particularly exposed to penalties for long work hours and motherhood, for example. In line with the findings by [Cha and Weeden \(2014\)](#), these penalties might be the result of a norm of overwork which leads to long work hours. Hence, regulations or changes to the taxation with regard to the compensation of long work hours might be considered to reduce gender inequality in the future. Policy measures might also be based on the results on heterogeneity by occupation and industry and focus on industries associated with large wage gaps. Another policy implication based on our findings could aim to improve women's access to higher education as we find that the extent of gender inequality is smaller for women with a bachelor's degree.

Finally, we would like to point at potential directions for future research. First of all, we consider our analysis as a first step to develop a more precise understanding of gender inequality in earnings. Future work might reproduce and reassess the reported heterogeneity patterns in other data sets, for example the Panel Study of Income Dynamics or the Current Population Survey and, of course, in other countries. We would like to particularly encourage the replication of our analysis in larger samples which might allow for more precise results. A point that we adopt from the recommendations of [Blau and Kahn \(2017\)](#) is that improved data quality on actual labour market experience, family interrelationships and information at the firm level would allow to get a better impression of the gender wage gap. Moreover, our study was restricted to full-time and year-round working employees. Thus, the extent to which women in our data set adjusted their labour market supply was restricted to the intensive margin, in other words how much hours they work each week in a full-time position (at least 35 hr). Future studies might analyse the gender gap heterogeneity in a yet broader sample including part-time working female workers, as well. Second, we would like to encourage additional work on causal mechanisms that drive the gender wage gap. Whereas our study is a first attempt describing heterogeneous patterns and, hence, reports association of variables and variation in the wage gap, future work on the underlying causal channels is of great relevance. We believe that this is particularly important in the context of the motherhood penalty which is likely to be driven by selection effects. For example, our results reflect the patterns in a larger gender wage gap for mothers of older children as compared to younger children

documented in Pal and Waldfogel (2016). We believe that causal evidence on the driving forces behind these patterns would contribute to develop a better understanding of gender inequality. Third, our methodology might be used in future to elabourate optimal equal pay policies. For instance, one could generate hypothetical quantile plots of the gender wage gap for different policy measures targeted at different subgroups and choose the policy that is optimal given a certain objective function and budget constraints.

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Conflict of interest: None declared.

Data availability

The code and preprocessed data files are available as supplementary materials. The original data files as downloaded from IPUMS-USA are available from <https://www.dropbox.com/sh/30sncv348o39m5u/AABMWybrDkOoqGdsG4QokT6Sa?dl=0> Please contact the corresponding author in case of questions or comments.

Supplementary material

[Supplementary material](#) is available online at *Journal of the Royal Statistical Society: Series A*.

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